

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Evaluation of the codebase's structure, readability, and reusability.

Code Layout Checklist

S.NO	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	Standard 4-space Python indentation
2	Proper File Structure	Files and folders are logically organized	Yes	app.py at root; templates/ and static/ follow Flask convention
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	temp, humidity, cloud, annual, janfeb, etc. map directly to form fields
4	Function / Method Names	Functions are descriptively named	Yes	home(), predict_page(), predict()
5	Code Comments	Inline and block comments explain logic where	Partial	Kept minimal by request;

		needed		logic is short and self-explanatory
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate routes for page rendering vs. prediction logic
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed an unused duplicate template (index.html) during cleanup
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Missing form fields currently raise a standard Flask 400; could add friendlier messaging

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	header_table() style navbar + theme toggle markup	HTML/CSS/JS	Repeated identically across home, predict, chance, no_chance templates	High
2	initTheme() / initRain() / initGauge()	JavaScript (main.js)	Shared across all pages via one main.js file	High
3	CSS design tokens (:root variables)	CSS3	Drives both dark and light themes from a single stylesheet	High

S.N o	Component / Module Name	Language / Technology	Where Reused	Reusability Level
4	Prediction DataFrame assembly block	Python (app.py)	Single predict() route; would be extracted into a helper if more model versions are added	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean separation of routes, templates, and static assets.
Readability	4	Clear naming; comments are intentionally minimal.
Reusability	4	Shared CSS/JS across pages; prediction logic could be further modularized.
Documentation / Comments	3	README and demo script cover usage; inline comments are sparse by design.
Overall Score	4	Solid, maintainable structure for the project's scope.